

Quantum eMotion Inc

Quantum Vacuum Random Number Generation — Cybersecurity Entropy at the Chip Level

PhD Due Diligence Report | BLRI-DD-QNC-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

True Randomness: The Foundation of All Cryptographic Security

Quantum eMotion builds Quantum Random Number Generators (QRNGs) using quantum vacuum fluctuations — a source of genuine randomness that cannot be predicted or replicated by classical deterministic algorithms. Cryptographic security depends entirely on unpredictable key generation: weak randomness is the root cause of billions in annual cybersecurity breaches. QNC's chip-scale QRNG enables quantum-certified entropy at mobile device, IoT, and server scale — an Ω -reduction technology that strengthens the cryptographic infrastructure underlying all of civilisational digital commerce and communication.

moomoo OpenD [2026-08-13]: Price \$2.475 | Mkt Cap \$542.9M | EPS TTM -\$0.039 | Net Income -\$8.6M | PE TTM -66.89x | 52w: \$0.693–\$4.56 | Shares 219.4M

Source: moomoo OpenD [2026-08-13] · Quantum eMotion investor relations · QRNG research

PART II — QUALITATIVE + QUANTITATIVE

Chip-Scale QRNG: Mobile and IoT Deployment

Quantum eMotion's key innovation is miniaturisation: its QRNG fits on a chip (vs rack-mounted systems from ID Quantique) enabling deployment in mobile phones, IoT sensors, and embedded systems. This is the product form factor that Quantinuum's IronBridge cannot serve (IronBridge is a cloud service, not

embedded hardware). The target: OEM integration in smartphone SoCs, automotive ECUs, and industrial PLCs.

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]

Price: \$2.475 | Mkt Cap: \$542.9M | EPS TTM: -\$0.039 | Net Income: -\$8.6M
PE TTM: -66.89x | 52w: \$0.693-\$4.56 | Shares: 219.4M | Canadian listed

PART III — COMPETITIVE + ETHNOGRAPHIC

Competitor	Type	vs QNC
ID Quantique	QRNG (rack-scale)	Larger; QNC chip-scale is smaller + cheaper for mobile
Quantinuum IronBridge	Cloud QRNG	Cloud vs chip; QNC for embedded deployments
Intel/ARM (TRNG)	Classical TRNG in silicon	Classical randomness; QNC is quantum-certified
SEALSQ (LAES)	PQC + QRNG planned	QNC pure QRNG specialist; LAES broader PQC

FINDING F-5-A: Portrait 1: The Mobile Security Architect (COMPOSITE)

A security engineer at a major smartphone OEM evaluates QNC's chip-scale QRNG for next-generation handset crypto acceleration. 'Classical TRNGs fail on mobile — predictable patterns allow key extraction attacks. QNC's quantum vacuum source is certified unbreakable.' Represents the smartphone OEM channel — a 1.4 billion unit annual market if QNC achieves SoC integration.

COMPOSITE / ILLUSTRATIVE — drawn from mobile security architecture patterns

FINDING F-5-B: Portrait 2: The Canadian Defence Contractor (COMPOSITE)

A software engineer at a Canadian aerospace company evaluates QNC QRNG for classified military communication encryption. 'Canadian-made QRNG for Canadian military comms — CCCS approved. QNC ticks every box for our procurement team.' Represents the defence market where Canadian-made quantum security hardware has preferred-vendor status.

COMPOSITE / ILLUSTRATIVE — drawn from Canadian defence procurement patterns

FINDING F-5-C: Portrait 3: The Bank Key Management Engineer (COMPOSITE)

A cryptographic infrastructure engineer at a G-SIB bank upgrades HSMs to quantum-safe key generation. 'We moved from NIST-approved DRBG to QNC QRNG for PCI-DSS compliance. The audit passed first time.' Represents financial services where QRNG is a compliance accelerator for PCI-DSS, SOX, and Basel cybersecurity requirements.

COMPOSITE / ILLUSTRATIVE — drawn from financial services cryptographic infrastructure patterns

PART IV — ADVERSARIAL + SUPERFORECASTING

HH1: HH1: ARM/Intel add QRNG to standard SoC architecture

EVIDENCE FOR:

If ARM includes quantum vacuum randomness source in Cortex-M standard; or Intel adds on-die photonic QRNG — QNC's addressable market collapses

EVIDENCE AGAINST:

SoC-level QRNG requires silicon photonic integration not standard in CMOS; QNC's stand-alone chip is cheaper than custom SoC modification; certification process favours dedicated QRNG vendors

QUANTITATIVE TEST

P(ARM/Intel include QRNG in standard SoC by 2028)=15%

VERDICT: BEAR tail risk — long-term threat but unlikely near-term

Implication: QNC must achieve OEM design-ins before SoC vendors build competitive capability

HH2: HH2: Mobile OEM design-in creates 100M+ unit volume

EVIDENCE FOR:

One smartphone OEM design-in creates enormous chip volume; QNC's unit economics improve dramatically at scale; revenue inflects from near-zero to \$20M+ in one contract

EVIDENCE AGAINST:

Smartphone OEM procurement cycles are 2-3 years minimum; QNC's chip must pass carrier certification and OEM security validation

QUANTITATIVE TEST

P(QNC achieves 1+ smartphone OEM design-in by 2028)=18%

VERDICT: BULL — mobile OEM is the transformative scenario

Implication: Monitor OEM partnership announcements; CTIA certification status

HH3: HH3: Classical TRNG sufficient; QRNG mandate never arrives

EVIDENCE FOR:

NIST's SP 800-90 approved DRBGs are sufficient for most cryptographic applications; no regulator mandates hardware QRNG; QNC's value proposition is 'nice to have' not 'must have'

EVIDENCE AGAINST:

Post-quantum transition makes key quality more critical; IronBridge's success validates QRNG demand in financial sector

QUANTITATIVE TEST

P(No regulatory QRNG mandate by 2030, limiting QNC's market to premium segment)=45%

VERDICT: BEAR-NEUTRAL — absence of mandate is the most likely scenario

Implication: QNC must pursue premium market (defence/finance) where QRNG is already valued

HH4: HH4: SEALSQ/BTQ integrate QNC QRNG — partnership or acquisition

EVIDENCE FOR:

SEALSQ's PQC chips would be strengthened by QNC QRNG integration; acquisition of QNC at \$542.9M would give SEALSQ complete PQC security stack

EVIDENCE AGAINST:

QNC's market cap may exceed SEALSQ's ability to acquire; partnership licensing is more likely than acquisition

QUANTITATIVE TEST

P(SEALSQ or BTQ acquires or partners with QNC by 2027)=20%

VERDICT: BULL — acquisition/partnership would validate QNC IP value

Implication: Monitor SEALSQ/BTQ partnership pipeline; QNC licensing activity

HH5: HH5: Pre-revenue small cap; stock volatility destroys retail holders

EVIDENCE FOR:

52w range \$0.693–\$4.56 = 559% volatility; retail-dominated; net loss -\$8.6M on near-zero revenue; any dilution is devastating

EVIDENCE AGAINST:

Small absolute loss (\$8.6M) gives longer runway than most peers; Canadian government grants provide non-dilutive support

QUANTITATIVE TEST

P(stock below \$1.50 within 12 months on sentiment collapse)=25%

VERDICT: BEAR — extreme volatility is structural for micro-cap pre-revenue

Implication: Position sizing essential; micro-cap liquidity risk is real

SCENARIO 1: Smartphone OEM design-in announced (P = 15%)

1+ OEM integrates QNC QRNG in mobile handset security chip; 100M unit volume potential; stock 3-5x on announcement

Driver: OEM partnership announcements; CTIA/GlobalPlatform certifications

Falsifier: OEM selects ARM TRNG or internal solution

SCENARIO 2: Canadian defence QRNG contract (P = 20%)

DND Canada or CSE awards QNC QRNG contract for classified comms; \$5M+ revenue; CCCS certification milestone

Driver: Defence procurement announcements; CCCS certification

Falsifier: Defence selects ID Quantique or imports from allied nation vendor

SCENARIO 3: Base case: certification + pilot customers (P = 38%%)

QNC achieves IC3 / NIST certification; 3-5 pilot customers in finance/defence; revenue \$1-3M; losses narrow; stock \$1.50-3.50

Driver: Certification announcements; pilot customer press releases

Falsifier: No certification by end 2026; stock falls on delay

SCENARIO 4: Financial institution QRNG adoption wave (P = 18%%)

5+ Tier 2 banks deploy QNC QRNG for key generation; PCI-DSS compliance catalyst; ARR \$5M+; first material revenue

Driver: Bank procurement announcements; PCI-DSS updated QRNG guidance

Falsifier: Banks adopt Quantinuum IronBridge cloud QRNG instead

SCENARIO 5: Stock falls to \$0.69 — near all-time low (P = 20%%)

Sentiment collapse + no revenue announcement; micro-cap selloff; near all-time low \$0.693 retested

Driver: Two consecutive empty quarters; macro quantum selloff

Falsifier: OEM design-in or defence contract announced

SCENARIO 6: SEALSQ or BTQ acquires QNC for QRNG IP (P = 12%%)

PQC company acquires QNC QRNG capability; acquisition at \$3-4/share (25-60% premium); IP integration

Driver: SEALSQ/BTQ partnership signals; QNC investor day

Falsifier: No acquisition interest; QNC remains independently loss-making

SCENARIO 7: EU eIDAS 2.0 mandates QRNG for national e-ID (P = 12%%)

EU eIDAS 2.0 update requires quantum-certified entropy for national digital identity; QNC chip-scale QRNG is the only certified embedded solution

Driver: eIDAS 2.0 technical standards committee updates

Falsifier: eIDAS deems classical TRNG sufficient for digital identity

SCENARIO 8: 5G authentication mandates QRNG (3GPP standard update) (P = 8%%)

3GPP releases 5G-Advanced security specification mandating QRNG for base stations; QNC telecom module approved; volume contract with Nokia/Ericsson

Driver: 3GPP meeting outcomes; 5G security specification updates

Falsifier: 3GPP does not mandate QRNG; classical entropy deemed sufficient

SCENARIO 9: Quantum computing awareness drives retail speculation (P = 25%%)

News cycle about quantum computing milestone drives retail buying; QNC stock trades to \$4+ on sentiment (not fundamentals); exit opportunity

Driver: Quantum computing news events; Reddit/social media sentiment

Falsifier: Retail interest moves to larger-cap quantum stocks

SCENARIO 10: Bull case: mass QRNG adoption — \$30M revenue by 2029 (P = 3%%)

OEM + defence + finance + IoT QRNG mandates all fire; QNC chips in 50M+ devices; profitable; stock \$8+

Driver: All four market channels adopt simultaneously

Falsifier: No mandate; classical TRNG remains the standard

Variable	QNC Impact	Direction	Magnitude
Ω	True QRNG eliminates predictable randomness — direct cryptographic Ω-reduction	↓↓ Strong	Ω-- foundational
I	Quantum-certified entropy strengthens institutional digital infrastructure	↑↑ High	I++ cryptographic
K	Secure randomness enables safe accumulation of K in digital form	↑ Enabling	K+ foundational
ξ	Chip-scale QRNG enables equal cryptographic security across device classes	↑ Moderate	ξ+ in digital equity

Risk	Probability	Impact	Mitigation
No QRNG mandate —	HIGH (45%)	HIGH	Defence + finance premium

voluntary market only			channels
ARM/Intel SoC QRNG integration	LOW (15%)	CRITICAL	Design-ins before SoC vendors move
Micro-cap volatility / liquidity risk	HIGH (60%)	MEDIUM	Position sizing; options unavailable
Pre-revenue dilution	MEDIUM (25%)	HIGH	Canadian government grants; certification milestones

CONVICTION VERDICT: HIGHLY SPECULATIVE — FOUNDATIONAL TECHNOLOGY, MICRO-CAP EXECUTION RISK

QNC at \$2.475 (mkt cap \$542.9M) has the most foundational Ω -reduction technology: true quantum randomness is the bedrock of all cryptographic security. The challenge: no regulatory mandate, pre-revenue, micro-cap with extreme volatility. Key catalyst: any OEM design-in or defence contract would re-rate 3-5 $\times$. HUMAN_ONLY_MANUAL. moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,492ch.